

Sarah C. Millholland

MIT Kavli Institute for Astrophysics and Space Research
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RESEARCH INTERESTS Exoplanet detection & characterization; demographics & orbital architectures of planetary systems; planetary dynamics and celestial mechanics; planetary atmospheres

EDUCATION

Yale University, New Haven, CT July 2016 – May 2020
Ph.D. in Astronomy, May 2020
Thesis: Data-Driven Dynamics of Planetary Systems
Advisor: Prof. Greg Laughlin
M.S., M.Phil. in Astronomy, May 2018

UC Santa Cruz, Santa Cruz, CA Sept. 2015 – June 2016
Pursuit of Ph.D. in Astronomy & Astrophysics
(transferred after completing first year)

University of Saint Thomas, Saint Paul, MN Sept. 2011 – May 2015
B.S. in Physics; **B.A.** in Mathematics, May 2015
Summa Cum Laude

POSITIONS

Assistant Professor July 2022 – present
Department of Physics, Massachusetts Institute of Technology

NASA Sagan Fellow July 2020 – June 2022
Department of Astrophysical Sciences, Princeton University
Supervisor: Prof. Joshua Winn

NSF Graduate Research Fellow 2017 – 2020
Department of Astronomy, Yale University
Advisor: Prof. Greg Laughlin

Graduate Student Researcher 2015 – 2017
Department of Astronomy, Yale University (2016 – 2017)
Department of Astronomy & Astrophysics, UCSC (2015 – 2016)
Advisor: Prof. Greg Laughlin

AWARDS

- UROP Outstanding Faculty Mentor Award 2025
- Sloan Research Fellowship 2025
- Vera Rubin Early Career Prize of the Division of Dynamical Astronomy 2024
- Brouwer Prize (“awarded to Yale astronomy graduates for contributions of unusual merit to any branch of astronomy”) 2021
- Lyman Spitzer Jr. Postdoctoral Fellowship, Princeton University 2020 – 2022
- NASA Hubble Fellowship Program (NHFP) Sagan Fellowship 2020 – 2022
- Tinsley Award (“best paper by a Yale astronomy graduate student”; for Millholland & Laughlin 2017b) 2018
- DDA/AAS Raynor L. Duncombe Prize for Student Research 2018
- Yale Conference Travel Fellowship 2017
- NSF Graduate Research Fellowship 2017 – 2020
- UCSC Regents Fellowship 2015
- NSF Graduate Research Fellowship Honorable Mention 2015

	- Barry M. Goldwater Scholarship (national science scholarship) 2014 – 2015 - Smith Academic Scholarship 2014 – 2015 - Danger Mathematics Scholarship 2013 – 2014, 2014 – 2015 - UST Collaborative Inquiry Research Scholarship 2014 - Walczak Mathematics Scholarship 2013 – 2014 - B. John Barry Academic Scholarship 2012 – 2013 - UST Endowed Scholarship 2011 – 2015
TEACHING	- Classical Mechanics III (8.09/8.309, graduate level) Spring 2024, Spring 2025 - Physics I: Classical Mechanics (8.012) Fall 2022, Fall 2023 - Physics II: Electricity and Magnetism (8.02) Spring 2023
ADVISING	Postdoctoral Researchers - Timothy Hallatt, MIT postdoctoral researcher, Fall 2024 – present Graduate students - Ritika Sethi, MIT graduate student, Fall 2023 – present - Emma Loudon, Yale graduate student, Fall 2022 – Spring 2025 - Anna Simpson, MIT graduate student, Fall 2023 – Spring 2024 Undergraduate students - Cyan Chandra, MIT, Spring 2026 – present - Nairi Davidian, MIT, Spring 2026 – present - Haedam Im, MIT, Fall 2022 – present - DJ Liveoak, MIT, Summer 2023 – present - Sidhant Suar, Fall 2023 – June 2025 - Brennen Black, MIT, Spring 2023 – Spring 2024 - Orion Foo, MIT, Fall 2022 – Fall 2023 - Teo Lara, MIT, Spring 2023 – Summer 2023 - Jan Toomlaid, MIT, Spring 2023 – Summer 2023 - Felicia Xiao, MIT, Summer 2023 - Nicole Sobski, Wellesley, Summer 2022 – Spring 2023 - Nicole Gountanis, Princeton, Fall 2021 - Joshua Zou, Princeton, Summer 2021 - David Jensen, Princeton, Fall 2020 - Samantha Berek, Yale, 2018 – 2020
REFEREED PUBLICATIONS	First author (underline=student supervised by S.C.M.) 17. Millholland, S. & Winn, J. “Exploring Exoplanet Dynamics with JWST: Tides, Rotation, Rings, and Moons.” 2025, PNAS, 122, 39, e2416189122 16. Millholland, S., MacLeod, M., & <u>Xiao, F.</u> “Empirical Constraints on Tidal Dissipation in Exoplanet Host Stars.” 2025, ApJ, 981, 77 15. Millholland, S., <u>Lara, T.</u>, & <u>Toomlaid, J.</u> “Spin Dynamics of Planets in Resonant Chains.” 2024, ApJ, 961, 203 14. Millholland, S., He, M., & Zink, J. “Edge-of-the-Multis: Evidence for a Transition in the Outer Architectures of Compact Multi-Planet Systems.” 2022, AJ, 164, 72 13. Millholland, S. & Winn, J. “Split Peas in a Pod: Intra-System Uniformity of Super-Earths and Sub-Neptunes.” 2021, ApJL, 920, L34

12. **Millholland, S.**, He, M., Ford, E., et al. “Evidence for a Non-Dichotomous Solution to the Kepler Dichotomy: Mutual Inclinations of Kepler Planetary Systems from Transit Duration Variations.” 2021, *AJ*, 162, 166
11. **Millholland, S.** & Spalding, C. “Formation of Ultra-Short-Period Planets by Obliquity-Driven Tidal Runaway.” 2020, *ApJ*, 905, 71
10. **Millholland, S.**, Petigura, E., & Batygin, K. “Tidal Inflation Reconciles Low-Density Sub-Saturns with Core Accretion.” 2020, *ApJ*, 897, 7
9. **Millholland, S.** “Tidally Induced Radius Inflation of Sub-Neptunes.” 2019, *ApJ*, 886, 72
8. **Millholland, S.** & Batygin, K. “Excitation of Planetary Obliquities Through Planet-Disk Interactions.” 2019, *ApJ*, 876, 119
7. **Millholland, S.** & Laughlin, G. “Obliquity-Driven Sculpting of Exoplanetary Systems.” 2019, *Nature Astronomy*, 3, 424
6. **Millholland, S.** & Laughlin, G. “Obliquity Tides May Drive WASP-12b’s Rapid Orbital Decay.” 2018, *ApJL*, 869, L15
5. **Millholland, S.**, Laughlin, G., Teske, J., et al. “New Constraints on Gliese 876 – Exemplar of Mean-Motion Resonance.” 2018, *AJ*, 155, 106
4. **Millholland, S.**, Wang, S., & Laughlin, G. “*Kepler* Multi-Planet Systems Exhibit Unexpected Intra-system Uniformity in Mass and Radius.” 2017, *ApJL*, 849, L33
3. **Millholland, S.** & Laughlin, G. “Supervised Learning Detection of Sixty Non-Transiting Hot Jupiter Candidates.” 2017, *AJ*, 154, 83
2. **Millholland, S.** & Laughlin, G. “Constraints on Planet Nine’s Orbit and Sky Position within a Framework of Mean-motion Resonances.” 2017, *AJ*, 153, 91
1. **Millholland, S.**, Wang, S., & Laughlin, G. “On the Detection of Non-Transiting Hot Jupiters in Multiple Planet Systems.” 2016, *ApJL*, 823, L7

Second author (underline=student supervised by S.C.M.)

16. Liveoak, D., **Millholland, S.**, Vick, M. “Case Study of a Super-eccentric Warm Jupiter Migrating via Equilibrium and Dynamical Tides.” 2026, *ApJ*, submitted
15. Lu, T., **Millholland, S.**, Rice, M., Black, B., et al. “The Architecture of the 14 Hercules System Suggests Primordial Ejection of a Massive Planet.” 2026, *ApJ*, submitted
14. Hallatt, T. & **Millholland, S.** “Shedding Light on Desert Dwellers.” 2026, *ApJ*, 997, 139
13. Hallatt, T. & **Millholland, S.** “Coupled Planetary Interior and Tidal Evolution.” 2026, *ApJ*, 997, 138
12. Liveoak, D., **Millholland, S.**, Vick, M., Tamayo, D. “Self-consistent Dynamical and Chaotic Tides in the REBOUNDx Framework.” 2025, *ApJ*, 989, 1, 35
11. Sethi, R. & **Millholland, S.** “Misaligned Planets Exhibit Greater Tidally Induced Radius Inflation Compared to Aligned Planets.” 2025, *ApJ*, 988, 247
10. Suar, S. & **Millholland, S.** “Planetary Obliquity Excitation Through Pre-Main Sequence Stellar Evolution.” 2025, *ApJ*, 989, 1, 111
9. Liveoak, D. & **Millholland, S.** “Formation of Close-in Neptunes Around Low-Mass Stars Through Breaking Resonant Chains.” 2024, *ApJ*, 974, 207
8. Louden, E. & **Millholland, S.** “Polar Neptunes are Stable to Tides.” 2024, *ApJ*, 974, 304
7. Gupta, A., **Millholland, S.**, Im, H., et al. “A hot Jupiter progenitor on a super-eccentric, retrograde orbit.” 2024, *Nature*, 632, 50

6. Sobski, N. & **Millholland, S.** “Can Cold Jupiters Sculpt the Edge-of-the-Multis?” 2023, *ApJ*, 954, 137
5. Weiss, L., **Millholland, S.**, Petigura, E., Adams, F., Batygin, K., Bloch, A., & Mordasini, C. “Architectures of Compact Multi-planet Systems: Diversity and Uniformity.” 2023, *Protostars and Planets VII* book chapter, arXiv: 2203.10076
4. Jensen, D. & **Millholland, S.** “Inferred Properties of Planets in Mean-Motion Resonances are Biased by Measurement Noise.” 2022, *AJ*, 164, 144
3. Lillo-Box, J., **Millholland, S.**, & Laughlin, G. “Follow-up of Non-Transiting Planets Detected by Kepler: Confirmation of Three Hot Jupiters and Validation of Three Other Planets.” 2021, *A&A*, 654, 9
2. Spalding, C. & **Millholland, S.** “Stellar Oblateness versus Distant Giants in Exciting *Kepler* Planet Mutual Inclinations.” 2020, *AJ*, 160, 105
1. Adams, A. D., **Millholland, S.**, & Laughlin, G. “Signatures of Obliquity in Thermal Phase Curves of Hot Jupiters.” 2019, *AJ*, 158, 3

Other co-author (underline=student supervised by S.C.M.)

30. Sethi, R., MacLeod, M., **Millholland, S.** “Atmospheric Escape Rates of Planets in Stellar Tidal Fields from 3-D Hydrodynamic Simulations.” 2026, *ApJ*, submitted
29. Garza, M., Limbach, M. A., Bowens-Rubin, R., et al. including **Millholland, S.** “Direct Imaging Constraints on Binary Planets and Exomoons around Epsilon Indi A b.” 2026, *AJ*, in press
28. Hallatt, T., Owen, J., & **Millholland, S.** “Revealing the Origin of Desert Dwellers via Stellar Obliquities.” 2026, *ApJL*, in press
27. Fitzmaurice, E., Stefánsson, G., Mahadevan, S., Im, H., **Millholland, S.**, et al. “An Orbital History and Dynamical Future for the Super-Eccentric Giant Planet Gaia 3b.” 2026, *AJ*, submitted
26. Liveoak, D., Hallatt, T., & **Millholland, S.** “Stability of Multiplanet Systems Through Hot Jupiter Destruction.” 2026, *ApJL*, 1002, 30
25. Mireles, I., Ulmer-Moll, S., Liveoak, D., Dragomir, D., Korth, J. et al. including **Millholland, S.** “Uncovering the Rapidly Evolving Orbits of the Dynamic TOI-201 System.” 2026, *Science Advances*, 12, eaef2618
24. Fernandes, Rachel B., Johnson, Samson, Bergsten, Galen J., et al. including **Millholland, S.** “Are We There Yet? Challenges in Quantifying the Frequency of Earth Analogs in the Habitable Zone.” 2025, *PASP*, 137, 12, 121001
23. Jenkins, S., Vanderburg, A., Sethi, R., **Millholland, S.**, et al. “An Eccentric Sub-Neptune Moving Into the Evaporation Desert.” 2025, *ApJL*, 989, 2, L20
22. Bardalez Gagliuffi, D., Balmer, W., Pueyo, L., Brandt, T., Giovannazzi, M., **Millholland, S.**, Black, B., et al. “JWST Coronagraphic Images of 14 Her c: A Cold Giant Planet in a Dynamically Hot, Multi-planet System.” 2025, *ApJL*, 988, 1, L18
21. Sun, M., Xie, J., Zhou, J., Liu, B., Nikolaou, N., **Millholland, S.** “Planetary Edge Trends (PET). I. The Inner Edge-Stellar Mass Correlation.” 2025, *A&A*, 699, A333
20. Lockley, I., Armstrong, D., Fernández Fernández, J., **Millholland, S.**, et al. “The TOI-1117 Multi-planetary System: 3 sub-Neptunes, 1 in both the Neptunian Desert and Radius Valley.” 2025, *MNRAS*, 541, 2, 919
19. Kunimoto, M., Zifan, L., **Millholland, S.**, et al. “Two Earth-size Planets and an Earth-size Candidate Transiting the Nearby Star HD 101581.” 2025, *AJ*, 169, 47

18. Lu, T., An, Q., Li, G., **Millholland, S.**, Brandt, G. M., Brandt, T. D. “Planet-Planet Scattering and ZLK Migration: The Dynamical History of HAT-P-11.” 2025, *ApJ*, 979, 218
17. Dai, F., Goldberg, M., Batygin, K., et al. including **Millholland, S.** [14 total] “The Prevalence of Resonance Among Young, Close-in Planets.” 2024, *AJ*, 168, 239
16. Burt, J., Hooton, M., Mamajek, E., Barragán, O., **Millholland, S.**, et al. [15 total] “TOI-1685 b is a Hot Rocky Super-Earth: Updates to the Stellar and Planet Parameters of a Popular JWST Cycle 2 Target.” 2024, *ApJL*, 971, L12
15. Leleu, A., Delisle, J., Burn, R., Izidoro, A., Udry, S., Dumusque, X., Lovis, C., **Millholland, S.**, et al. [15 total] “Resonant Sub-Neptunes are Puffier.” 2024, *A&A*, 687, L1
14. Burdge, K., El-Badry, K., Kara, E., Canizares, C., Chakrabarty, D., Frebel, A., **Millholland, S.**, Rappaport, S., Simcoe, R., Vanderburg, A. “The black hole low mass X-ray binary V404 Cygni is part of a wide hierarchical triple, and formed without a kick.” 2024, *Nature*, 635, 316
13. Fairnington, T., Nabbie, E., Huang, C., Zhou, G., Foo, O., **Millholland, S.**, et al. “TOI-5126: A hot super-Neptune and warm Neptune pair discovered by TESS and CHEOPS.” 2024, *MNRAS*, 527, 8768
12. Louden, E., Laughlin, G., & **Millholland, S.** “Evidence for Two Dissipation Regimes Among the Short-Period Exoplanets.” 2023, *ApJL*, 958, L21
11. Murgas, F., Castro-González, A., Pallé, Pozuelos, F., **Millholland, S.**, Foo, O., et al. “Two super-Earths at the edge of the habitable zone of the nearby M dwarf TOI-2095.” 2023, *A&A*, 677, A182
10. Lu, T., Rein, H., Tamayo, D., Hadden, S., Mardling, R., **Millholland, S.**, & Laughlin, G. “Self-Consistent Spin, Tidal and Dynamical Equations of Motion in the REBOUNDx Framework.” 2023, *ApJ*, 948, 41
9. Bozhilov, V., Antonova, D., Hobson, M., Brahm, R., Jordán, A., et al. including **Millholland, S.** “A 2:1 Mean-Motion Resonance Super-Jovian Pair Revealed by TESS, FEROS, and HARPS.” 2023, *ApJL*, 946, L36
8. Zhao, L., Kunovac, V., Brewer, J., Llama, J., **Millholland, S.**, et al. [11 total] “Measured Spin-Orbit Alignment of Ultra-Short Period Super-Earth 55 Cancri e.” 2023, *Nature Astronomy*, 7, 198
7. Stefansson, G., Mahadevan, S., Petrovich, C., Winn, J., Kanodia, S., **Millholland, S.**, et al. [37 total] “The Warm Neptune GJ 3470b has a Polar Orbit.” 2022, *ApJL*, 931, L15
6. Azari, A., Biersteker, J., Dewey, R., Doran, G., et al. including **Millholland, S.** [16 total] “Integrating Machine Learning for Planetary Science: Perspectives for the Next Decade.” 2021, *BAAS*, 53, 128 (white paper)
5. Davis, A., Wang, S., Jones, M., Eastman, J., Günther, M., Stassun, K., et al. including **Millholland, S.** [51 total] “TOI 564 b and TOI 905 b: Grazing and Fully Transiting Hot Jupiters Discovered by TESS.” 2020, *AJ*, 160, 229
4. Bryan, M., Chiang, E., Bowler, B. P., Morley, C. V., **Millholland, S.**, Blunt, S., Ashok, K. B., Nielsen, E., Ngo, H., Mawet, D., Knutson, H. A. “Obliquity Constraints on an Extrasolar Planetary-Mass Companion.” 2020, *AJ*, 159, 181
3. Wang, S., Jones, M., Shporer, A., Fulton, B. J., Paredes, L. A., Trifonov, T., Kossakowski, D., Eastman, J., Redfield, S., Günther, M. N., Kreidberg, L., Huang, C. X., **Millholland, S.**, et al. [60 total] “HD 202772Ab: A Transiting Hot Jupiter Around a Bright, Mildly Evolved Star in a Visual Binary Discovered by TESS.” 2019, *AJ*, 157, 51

2. Becker, J. C., Khain, T., Hamilton, S. J., Adams, F. C., Gerdes, D. W., Zullo, L., Franson, K., **Millholland, S.**, et al. [66 total] “Discovery and Dynamical Analysis of an Extreme Trans-Neptunian Object with a High Orbital Inclination.” 2018, AJ, 156, 81
1. Janvier, M., Savcheva, A., Pariat, E., Tassev, S., **Millholland, S.**, Bommier, V., McCauley, P., McKillop, S., Dougan, F. “Evolution of Flare Ribbons, Electric Currents and Quasi-separatrix Layers During an X-class Flare.” 2016, A&A, 591, A141

SEMINARS &
COLLOQUIA
(* = INVITED)

- * Astronomy Colloquium, Carnegie Observatories, January 2026
- * Physics & Astronomy Colloquium, Colby College, December 2025
- * Astronomy Colloquium, University of Toronto, January 2025
- * Astrophysics Seminar, Institute for Advanced Study, November 2024
- * Astrophysics Seminar, Southwest Research Institute, October 2024
- * Five College Astronomy Department Colloquium, University of Massachusetts Amherst, March 2024
- * Physics Colloquium, Clark University, March 2024
- * Exoplanet Seminar, University of Geneva, Sept. 2023
- * Astronomy Colloquium, The Ohio State University, April 2023
- * Physics & Applied Math Seminar, University of St. Thomas, March 2023
- * Institute for Theory and Computation Colloquium, Center for Astrophysics — Harvard & Smithsonian, Nov. 2022
- * Planetary Lunch Seminar, Massachusetts Institute of Technology, Sept. 2022
- * Astrophysics Colloquium, Chalmers University of Technology, May 2022
- * Astronomy Colloquium, University of Wisconsin-Madison, March 2022
- * Stellar Astrophysics Center Seminar, Aarhus University, November 2021
- * Astronomy Colloquium, California Institute of Technology, October 2021
- * Astronomy Colloquium, Columbia University, October 2021
- * Astronomy Colloquium, University of Rochester, September 2021
- * Astrophysics Seminar, Astrophysical Fluid Dynamics Group at DAMTP, University of Cambridge, May 2021
- * Astronomy Colloquium, University of California Los Angeles, March 2021
- * Center for Integrative Planetary Science Seminar, University of California Berkeley, March 2021
- * Exoplanets and Protoplanetary Disks Research Group Talk, Imperial College London, March 2021
- * Canadian Institute for Theoretical Astrophysics Seminar, University of Toronto, March 2021
- * Center for Astrophysical Sciences Seminar, Johns Hopkins University, Feb. 2021
- * Astronomy Seminar, Carnegie Earth and Planets Laboratory, Feb. 2021
- * Astrophysics Colloquium, Massachusetts Institute of Technology, Feb. 2021
- * Colloquium, Center for Computational Astrophysics, Flatiron Institute, Oct. 2020
- * JILA Astrophysics Seminar, University of Colorado Boulder, Oct. 2020
- * JPL Exoplanet Journal Club Talk, NASA Jet Propulsion Laboratory, Aug. 2020
- * Special Seminar, Climate and Space Sciences and Engineering, University of Michigan, Feb. 2020
- * Cosmos Seminar, The University of Texas at Austin, Oct. 2019
- * Institute for Theory and Computation Seminar, Center for Astrophysics — Harvard & Smithsonian, Oct. 2019
- Friday Lunch Time Astrophysics Seminar, University of California Santa Cruz, Oct.

2019

- Tuesday Lunch Talk, University of California Los Angeles, Oct. 2019
- Astronomy Tea Talk, California Institute of Technology, Oct. 2019
- Exoplanet Tea Talk, Massachusetts Institute of Technology, Sept. 2019
- * Planetary Lunch Seminar, Cornell University, Sept. 2019
- Astronomy Seminar, Columbia University, Sept. 2019
- Star and Planet Formation Seminar, University of Michigan, March 2019
- Exoplanet Lunch Seminar, Princeton University, Feb. 2019
- * Center for Exoplanets & Habitable Worlds Seminar, Penn State University, Feb. 2019
- * Extrasolar Planets Seminar, NASA Goddard Space Flight Center, April 2018
- * Planetary Science Seminar, California Institute of Technology, Dec. 2017
- * Stars & Planets Seminar, Harvard-Smithsonian Center for Astrophysics, Nov. 2017
- Exoplanet Pizza Lunch, Harvard-Smithsonian Center for Astrophysics, March 2017

CONFERENCE
TALKS

(* = INVITED)

- * Vera Rubin Early Career Prize Lecture, Division of Dynamical Astronomy Meeting, Chicago, IL, June 2026
- Division of Dynamical Astronomy Meeting, Atlanta, GA, May 2025
- * Invited talk, 27th International Microlensing Conference, Cape Town, South Africa, January 2025
- Exoplanets V, Leiden, Netherlands, June 2024
- * Invited talk, NASA SEEC Symposium: Pathways to Characterizing Non-Transiting Planets, NASA GSFC, April 2024
- Division of Dynamical Astronomy Meeting, University of Toronto, Toronto, ON, May 2024
- Extreme Solar Systems V, Christchurch, New Zealand, March 2024
- * Invited review, Division of Dynamical Astronomy Meeting, East Lansing, MI, May 2023
- * Invited review, Protostars & Planets VII (review of PPVII Chapter 24), Kyoto, Japan, April 2023
- AAS Meeting #241, Seattle, WA, January 2023
- Exoplanets IV, Las Vegas, NV, May 2022
- Division of Dynamical Astronomy Meeting, Flatiron Institute, New York, NY, April 2022
- 2021 NHFP Symposium, virtual conference, October 2021
- AAS Meeting #238, virtual conference, June 2021
- Division of Dynamical Astronomy Meeting, virtual conference, May 2021
- PLATO ESP Workshop on Planetary Interiors and System Architectures, virtual conference, November 2020
- 2020 NHFP Symposium, virtual conference, September 2020
- Division of Dynamical Astronomy Meeting, virtual conference, August 2020
- Boston Area Exoplanet Science Meeting #7, virtual conference, April 2020
- AAS Meeting #235, Honolulu, HI, January 2020
- Extreme Solar Systems IV, Reykjavik, Iceland, August 2019
- Emerging Researchers in Exoplanet Science (ERES) V, Cornell University, Ithaca, NY, June 2019
- Division of Dynamical Astronomy Meeting, Boulder, CO, June 2019
- Kepler & K2 Science Conference V, Glendale, CA, March 2019
- Boston Area Exoplanet Science Meeting, Boston University, Boston, MA, Jan. 2019
- 2018 CT Exoplanets Meeting, Wesleyan University, Middletown, CT, July 2018

- Emerging Researchers in Exoplanet Science (ERES) IV, Pennsylvania State University, State College, PA, June 2018
- Planet Nine Workshop, California Institute of Technology, Pasadena, CA, May 2018
- Division of Dynamical Astronomy Meeting, San Jose, CA, April 2018
- Numerical Integration Methods in Planetary Science, University of Toronto at Scarborough, Toronto, Ontario, August 2017
- Numerical Integration Methods in Planetary Science, University of Toronto at Scarborough, Toronto, Ontario, August 2017
- Kepler & K2 Science Conference IV, NASA Ames Research Center, Moffett Field, CA, June 2017
- Emerging Researchers in Exoplanet Science (ERES) III, Yale University, New Haven, CT, June 2017
- 2017 CT Exoplanets Meeting, Wesleyan University, Middletown, CT, May 2017
- 2017 Aspen Winter Conference, Formation and Dynamical Evolution of Exoplanets, Aspen, CO, March 2017

WEB BLOG PUBLICATIONS - “Tilting Planets and Sculpting Orbits”, invited guest post by S. Millholland at *Nature Research Behind the Paper*, March 2019

PROFESSIONAL ACTIVITIES & SERVICE **Reviews**

- Reviewer for Nature, Nature Astronomy, AJ, ApJ, ApJL, PSJ, Icarus, A&A, MNRAS, PNAS, PASP
- Panel reviewer, NSF (AAG, CAREER)
- Panel reviewer, NASA (XRP, TESS GI, HST TAC)
- External reviewer, Canada-France-Hawaii Telescope
- External reviewer, OPTICON Telescope Transnational Access

Organizing Committees

- Executive Committee, Division of Dynamical Astronomy 2021 – 2024
- SOC, DDA 56th Annual Meeting 2024 – 2025
- SOC, DDA 55th Annual Meeting 2023 – 2024
- SOC, Exoplanets V 2023 – 2024
- SOC, Extreme Solar Systems V 2023 – 2024
- SOC, DDA 54th Annual Meeting 2022 – 2023
- SOC, DDA 53rd Annual Meeting 2021 – 2022
- Organizer, Princeton Planetary Dynamics Reading Group 2020 – 2022
- SOC, Emerging Researchers in Exoplanet Science (ERES) Conference III (Yale), V (Cornell) and VI (virtual) 2017, 2019, 2021

MIT Physics Departmental Service

- Physics Values Committee 2023 – present
- Graduate Admissions Committee 2024
- MSRP Admissions Committee 2023
- Grader for Classical Mechanics Written Exam 2023, 2024
- Physics Core Requirements Review Ad Hoc Committee 2023
- Kavli Postdoctoral Fellowship Review Committee 2023
- Torres Postdoctoral Fellowship Review Committee 2023
- 51 Pegasi b Fellowship Review Committee 2023, 2024
- Organizing Committee, Prospective Graduate Student Open House 2023
- Astro Oral Exam Committee 2023

Diversity & Inclusion Involvement

- Committee member, MIT Physics Values Committee 2023 –
- Co-organizer, Seminar Series on Equity, Diversity and Inclusion for the Departments of Physics and Astrophysics at Princeton 2020 – 2021
- Committee member, Improving equity and inclusion in graduate admissions at the Department of Astrophysics at Princeton 2020
- Organizer, Panel Discussion at Emerging Researchers in Exoplanet Science V: “Diversity & Inclusion in Astronomy” 2019

Professional Affiliations & Mentorship

- Member, American Astronomical Society
- Member, Division of Dynamical Astronomy of the AAS
- Mentor, Goldwater Scholar Community Mentorship Program 2021 – 2023
- Mentor, DDA Mentorship Program 2021 – 2023
- Mentor, Princeton Astrophysics Mentorship Program 2021 – 2022

PUBLIC TALKS AND ARTICLES

- Amateur Telescope Makers of Boston, November 2024 (talk)
- Skyscrapers Astronomical Society of Rhode Island, August 2023 (talk)
- Warrior-Scholar Project, MIT, July & August 2023 (lectures)
- Assembly Speaker Series, Commonwealth School, April 2023 (talk)
- STEAM Speaker Series, International School of Boston, March 2023 (talk)
- Astronomy Club, International School of Boston, Jan. 2023 (talk)
- Aldrich Astronomical Society Meeting, Dec. 2022 (talk)
- Yale Young Global Scholars Research Showcase (<http://globalscholars.yale.edu>), Yale University, June & July 2022 (virtual talks)
- Exploring Science Program, Yale University, August 2020 (virtual talk)
- Yale Young Global Scholars Research Showcase, Yale University, July 2019 (talk)
- Institute for Learning in Retirement, Albertus Magnus College, October 2018 (talk)
- Yale Young Global Scholars Research Showcase, Yale University, June & July 2018 (talks)
- Leitner Family Observatory & Planetarium, February 2018 (talk)
- Pathways Summer Scholars Program Science Café (<http://pathwayssummerscholars.yale.edu>), Yale University, July 2017 (talk)
- Yale Young Global Scholars Research Showcase, Yale University, July 2017 (talk)
- Pathways Summer Scholars Enrichment Workshop, Yale University, July 2017 (talk & workshop)
- Yale Open Labs Science Café (<http://theopenlabs.org>), Yale University, April 2017 (talk)
- 9th Annual Women in Leadership Conference, Yale University, February 2017 (invited panel member)
- “The Search for Planet Nine”, a publication for the Hartford Courant News in Education series, *Science Matters!*, January 2017 (article)
- LAMAT REU Program, UC Santa Cruz, July 2016 (talk)
- “Reading Scientific Literature”, LAMAT REU Program, UC Santa Cruz, June 2016 (workshop)
- Public Observing Night, University of St. Thomas Observatory, March 2013 (talk & public observing)
- Minnesota Optical Society Meeting, March 2013 (talk)

SELECTED OUTREACH

- Over 20 public talks to date (see list above)

- Physics lecturer, Warrior-Scholar Project 2023
(<https://www.warrior-scholar.org/>)
- Guest speaker, MIT Physics First-Year Graduate Student Seminar 2023
- Yale Open Labs (<http://theopenlabs.org>) 2016 – 2018
Committee chair for Science Café Talk Series 2017 – 2018
Executive board member 2016 – 2017
- Organizer of Yale Young Global Scholars Program visits 2017, 2018
to the Astro. Department (<http://globalscholars.yale.edu>)
- Public Night Volunteer, Lick Observatory Summer 2016
- Astronomy Public Night Leader, UST Observatory 2012 – 2015

SELECTED
MEDIA
COVERAGE

- **Hot Jupiter progenitor** (Gupta, Millholland, et al. 2024) featured in EarthSky, Phys.org, MIT News, and others.
- **Tilted planets** (Millholland & Laughlin 2019) featured in Scientific American, Sky & Telescope, Popular Science, Science Daily, Live Science, Space.com, Yale News.
- **Intra-system uniformity** (Millholland et al. 2017) featured in AAS Nova, Nature Research Highlights.
- **Machine learning detection of hot Jupiters** (Millholland & Laughlin 2017b) featured in National Geographic, Sky & Telescope, FOX 61 Connecticut News (television), University of St. Thomas News, Yale News.
- **Featured subject** for University of St. Thomas television and online advertisements